UTA018

Roll Number:

Thapar Institute of Engineering and Technology, Patiala
Department of Computer Science

BE: Summer Semester
MST
1st July, 2026
Time: 2 Hours; M. Marks: *30*

Course Name: Object Oriented Programming

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Instructions:

1. All questions are compulsory.
2. All sub questions carry equal weightage.
3. Show all necessary steps for full credit.
4. Page number of the attempted question should be mentioned on the first page of the answer sheet.

Q.No.	Question	Marks	CO	BL
Q1	a) A university administration wants to know how many student records have been created in the system at any point in time. Design and implement a suitable C++ class to fulfill this requirement. Demonstrate the working of your program by creating multiple student objects.	(3)	CO1	L3
	b) Two statements are available for storing employee names given as: char name[20]; and char *name = new char[n]; Which approach is suitable under what circumstances? Justify your answer.	(1.5) (1.5)		

	c) Predict the output of given code: #include<iostream>
	using namespace std;
	class Test{
	public:
	Test(){
	cout<<"C ";}
	~Test(){
	cout<<"D ";}};
	void fun(){ static Test t;}
	int main(){ fun(); fun(); cout<<"M "; }			

Q2	a) The following constructor does not initialize the data members correctly: Employee(int id, float salary){
	id = id;
	salary = salary;
	} Explain the problem. Rewrite the constructor so that the object is initialized correctly.
		(1.5)	CO1	L5

	b) A hotel reservation system creates a room object only when a customer books a room. Write a suitable C++ program to model this requirement.			

	c) Identify the error in the given code and reason.
	class Test{
	public:
	int x=10;
	void modify(){x=20;}};
	int main(){
	const Test t;
	t.modify();}
		(3)
		
		(1.5)		

Q3	a) Consider the following classes:
	class Person
	{
	Public:
	Person(int age)
	{
	cout << "Person";
	}
	};
	class Student : public Person
	{
	public:
	Student()
	{
	cout << "Student";
	}
	};
	Does the following program generate a compilation error? If yes, identify the reason, write the corrected program, and justify why the correction is necessary.
	
	b) A programmer uses the default copy constructor in a class that allocates memory dynamically using new. Predict the problems that may occur during program execution and explain how a user-defined copy constructor solves them.
	c) Identify the error in the following code and justify your answer:
	#include<iostream>
	using namespace std;
	class Sample{
	public:
	void show() const {
	x=10;
	}
	private:
	int x;
	};
		(3)
		
		(1.5)
		
		(1.5)
			CO2	L3

Q4	a) Does the following program generate any runtime problem? If yes, identify the reason and rewrite the program using a user-defined copy constructor.
	#include<iostream>
	using namespace std;
	class Sample
	{
	int *x;
	public:
	Sample(int a)
	{
	x = new int(a);
	}
	~Sample()
	{
	delete x;
	}
	};
	int main()
	{
	Sample s1(10);
	Sample s2 = s1;
	}
		(1.5)	CO2	L4

	b) Create two classes **Student** and **Scholarship**. Declare **Scholarship** as a friend class of **Student**. Also, declare a **friend function** common to both classes. The friend class should determine scholarship eligibility, while the friend function should calculate the scholarship amount and display the student's final academic report. Use only private data members. Do not use getter functions. Initialize data using `input()` member functions.	(3)		

	c) Predict the output of the following code:
	#include<iostream>
	using namespace std;
	class Base{
	public:
	Base()
	{
	Cout<<"B ";
	}
	};
	class Derived: public Base
	{
	Public:
		(1.5)		

	Derived()
	{
	cout<<"D";
	}
	};
	Int main()
	{
	Derived d;
	}			

Q5	a) #include<iostream> using namespace std;
	class Base{
	public:
	void show(){
	cout<<"Base";}};
	class Derived : public Base{
	public:
	void show(){
	cout<<"Derived";
	}};
	int main(){ Base *ptr; Derived d; ptr = &d; ptr->show();
	}
	Predict the output of given code. Why is Derived::show() not called? **Write a modified version to invoke the derived class function.**
	
	a) Create a class PRODUCT containing Product ID and Product Name. Derive a class DISCOUNTED_PRODUCT containing Discount Percentage. Compute the final selling price after discount.
		(3)
		
		(3)
			CO2	L5